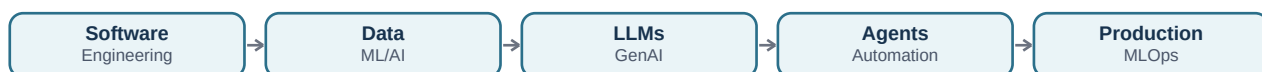


AI Engineering Interview Questions & Answers

Simple English • Short but Meaningful Answers • Practical Examples • Visual Learning



Designed for interview preparation: understand the concept, explain it simply, and connect it to a real production example.

Audio note: This PDF contains visual learning content. Interactive audio playback is not embedded because standard PDF audio support is not reliably supported across common PDF readers.

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How to Use This Guide in an Interview

- **Start with a definition:** explain the concept in one or two simple sentences.
- **Explain why it is used:** connect the technology to a real problem.
- **Give one example:** show that you understand practical usage.
- **Discuss trade-offs:** mention cost, latency, scalability, accuracy, security, or complexity when relevant.
- **Connect the system:** explain how the technology fits into the complete AI application.

A strong AI Engineer is not only someone who knows models. A strong AI Engineer can design, build, integrate, deploy, monitor, and improve reliable AI systems.

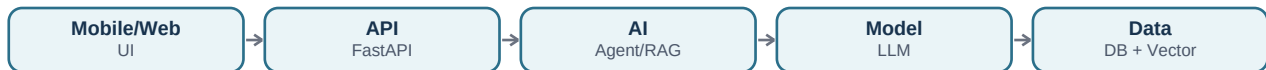
1. How would you design an AI application for mobile and web from start to end using SDLC?

Short Answer: I would follow a normal SDLC, but add AI-specific steps such as data, model evaluation, prompt testing, safety, and monitoring.

Key Points:

- 1) **Requirement:** Define the user problem, AI use case, success metrics, privacy, cost, and latency targets.
- 2) **Architecture:** Choose mobile/web clients, backend APIs, model provider or self-hosted model, database, vector database, and observability.
- 3) **Data:** Collect, clean, label, secure, and version training or knowledge data.
- 4) **Prototype:** Build a small proof of concept and test the AI quality.
- 5) **Development:** Build the backend, AI pipeline, prompts, tools, authentication, and mobile/web UI.
- 6) **Testing:** Unit, integration, security, load, prompt, hallucination, bias, and model-quality tests.
- 7) **Deployment:** CI/CD, Docker, cloud infrastructure, model versioning, and rollback.
- 8) **Monitoring:** Latency, errors, cost, token usage, user feedback, quality, and model drift.
- 9) **Improvement:** Analyze failures, improve data/prompts/models, and repeat the cycle.

Example: Example: A mobile AI assistant can use Flutter → FastAPI → Agent/RAG service → LLM → PostgreSQL + Vector DB. The mobile app should not directly expose secret model API keys.



2. What is Machine Learning (ML)?

Short Answer: Machine Learning is a way to make computers learn patterns from data and make predictions or decisions without writing every rule manually.

Key Points:

- **Input:** historical data.
- **Training:** the model learns patterns.
- **Inference:** the trained model predicts a new result.
- **Evaluation:** measure accuracy and other metrics.

Example: *Example: Give a model thousands of labeled emails. It learns to predict whether a new email is spam.*

3. What is Deep Learning?

Short Answer: Deep Learning is Machine Learning using neural networks with many layers. It is especially powerful for images, audio, language, and complex patterns.

Key Points:

- It learns features automatically.
- It normally needs more data and computing power than traditional ML.
- Popular frameworks include PyTorch and TensorFlow.

Example: *Example: A deep neural network can learn edges → shapes → objects to recognize a face in an image.*

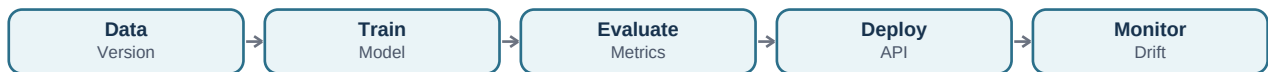
4. What is MLOps?

Short Answer: MLOps applies DevOps practices to Machine Learning. It manages the complete ML lifecycle: data, training, models, deployment, monitoring, and retraining.

Key Points:

- **Typical flow:** Data → Train → Evaluate → Register → Deploy → Monitor → Retrain.
- It improves reproducibility, automation, collaboration, and production reliability.

Example: Example: If model accuracy falls because customer behavior changes, MLOps detects the problem and triggers a new training cycle.



5. What is LLMOps?

Short Answer: LLMOps is the production discipline for Large Language Models. It manages prompts, models, RAG data, evaluations, token costs, latency, safety, and observability.

Key Points:

- Prompt/version management
- LLM and embedding model evaluation
- Token and cost tracking
- RAG quality monitoring
- Guardrails and security
- Model/provider fallback

Example: Example: An AI support bot may use one model for normal questions and automatically switch to another model when the first provider is unavailable.

6. What is n8n and how is it useful in AI?

Short Answer: n8n is a workflow automation platform. It connects APIs, databases, SaaS tools, webhooks, and AI services using visual workflows.

Key Points:

- Useful for AI automation, scheduled jobs, webhooks, document processing, notifications, and integrations.
- It can call REST APIs, databases, LLMs, and external tools.
- It is useful for orchestration, but complex core business logic may still belong in a proper backend service.

Example: Example: New PDF → extract text → create embeddings → store in Vector DB → send Slack notification.

7. What is MLflow?

Short Answer: MLflow is an open-source platform for managing the ML lifecycle. It can track experiments, parameters, metrics, artifacts, and model versions.

Key Points:

- **Tracking:** record experiments.
- **Registry:** manage model versions.
- **Deployment:** package and serve models.
- **Reproducibility:** know how a model was created.

Example: *Example: Compare three training runs and identify which model achieved the best validation score.*

8. What is Model Drift?

Short Answer: Model Drift happens when real-world data or behavior changes, causing a model to perform worse over time.

Key Points:

- **Data drift:** input data distribution changes.
- **Concept drift:** the relationship between input and output changes.
- **Prediction drift:** model output distribution changes.

Example: *Example: A fraud model trained on old shopping behavior may become less accurate when criminals change their methods.*

9. What is a Transformer?

Short Answer: A Transformer is a neural network architecture that uses attention to understand relationships between tokens. It is the foundation of most modern LLMs.

Key Points:

- **Attention:** identifies which tokens are important to each other.
- **Parallel processing:** handles sequences efficiently during training.
- **Examples:** GPT, BERT, T5, and many modern language models.

Example: Example: In “The animal did not cross the road because it was tired,” attention helps understand what “it” refers to.



10. What is DSPy?

Short Answer: DSPy is a framework for programming and optimizing LLM applications using structured modules and measurable evaluation instead of manually writing every prompt.

Key Points:

- Define the task and signature.
- Create modules such as prediction or reasoning.
- Evaluate results with a metric.
- Optimize prompts or demonstrations automatically.

Example: Example: Instead of manually trying 20 prompts for a classification task, DSPy can optimize the program against an evaluation metric.

11. What is gRPC?

Short Answer: gRPC is a high-performance RPC framework commonly used for communication between backend services. It usually uses Protocol Buffers and HTTP/2.

Key Points:

- Strongly typed contracts
- Fast binary serialization
- Streaming support
- Good for internal microservices

Example: *Example: An API Gateway calls an internal Python inference service using gRPC for low-latency communication.*

12. What is PyTorch?

Short Answer: PyTorch is a popular open-source deep learning framework used to build, train, fine-tune, and run neural networks.

Key Points:

- Dynamic computation graphs
- GPU acceleration
- Large research and production ecosystem
- Widely used in modern AI and LLM development

Example: *Example: Train a neural network, fine-tune a language model, or build a computer vision model using PyTorch.*

13. What is Hugging Face?

Short Answer: Hugging Face is an AI ecosystem that provides models, datasets, libraries, and tools for Natural Language Processing, computer vision, audio, and generative AI.

Key Points:

- Model Hub
- Transformers library
- Datasets
- Tokenizers
- Inference and deployment tools

Example: *Example: Download a pre-trained text classification model, fine-tune it, and deploy it behind an API.*

14. What is TensorFlow?

Short Answer: TensorFlow is an open-source machine learning framework used to build, train, and deploy ML and deep learning models.

Key Points:

- Works across CPUs, GPUs, and specialized hardware.
- Useful for training and production deployment.
- Includes tools such as Keras for easier model development.

Example: *Example: Build an image classification model and deploy it to a cloud service or mobile device.*

15. What is Clean Architecture and what are its layers?

Short Answer: Clean Architecture separates business rules from frameworks, databases, and UI. The main goal is low coupling and easier testing.

Key Points:

- **Entities/Domain:** core business rules.
- **Use Cases:** application actions.
- **Interface Adapters:** controllers, presenters, repositories.
- **Frameworks/Infrastructure:** database, API, UI, external services.

Example: Example: The business rule “user can only book an available slot” should not depend directly on Flutter or PostgreSQL.



16. What is MVVM?

Short Answer: MVVM means Model–View–ViewModel. It separates UI from presentation logic and data handling.

Key Points:

- **Model:** data and business models.
- **View:** screen/UI.
- **ViewModel:** state and presentation logic.
- It improves testability and maintainability.

Example: Example: In Flutter, a screen observes a *ViewModel* state such as loading, success, or error.

17. What are Microservices?

Short Answer: Microservices split a large application into small, independently deployable services. Each service owns a focused business capability.

Key Points:

- Independent deployment
- Service-specific scaling
- Technology flexibility
- Requires good networking, monitoring, security, and distributed-system design

Example: Example: User Service, Payment Service, AI Inference Service, Notification Service, and Search Service communicate through APIs or events.

18. What is Monolith Architecture?

Short Answer: A monolith is an application deployed as one main unit. Its features usually share the same codebase and deployment process.

Key Points:

- **Advantages:** simple development, testing, and deployment.
- **Disadvantages:** large codebase, coupled releases, and scaling may be less targeted.

Example: *Example: A single backend contains authentication, users, payments, and AI features in one deployable application.*

19. What is Redis?

Short Answer: Redis is an in-memory data store commonly used for caching, sessions, queues, rate limiting, and fast temporary data.

Key Points:

- Very fast key-value access
- TTL/expiration support
- Pub/Sub and streams
- Useful for caching expensive AI responses

Example: *Example: Cache the result of a common AI question for 10 minutes to reduce latency and LLM cost.*

20. Why is Python widely used in AI, and what can it do?

Short Answer: Python is popular in AI because it is easy to read and has a very large ecosystem for data, ML, deep learning, LLMs, APIs, automation, and deployment.

Key Points:

- **AI/ML:** scikit-learn, XGBoost, PyTorch, TensorFlow.
- **LLMs:** Transformers, LangChain, LangGraph, LlamaIndex, DSPy, AutoGen, CrewAI.
- **RAG:** document loaders, chunking, embeddings, reranking, Vector DB integration.
- **Backend:** FastAPI, Django, Flask, REST APIs, WebSockets, gRPC.
- **Data:** NumPy, pandas, Polars, SQL, PostgreSQL.
- **Cloud/Deployment:** Docker, Kubernetes, AWS, GCP, CI/CD.
- **Automation:** scripts, ETL, webhooks, scheduled jobs, n8n integration.
- **Mobile/Web:** Python normally provides the AI/backend API; Flutter, React, React Native, or native mobile apps provide the client UI.
- **Typical full stack:** Flutter/React → FastAPI → AI orchestration → LLM/model → PostgreSQL/Redis/Vector DB → Docker → AWS/GCP.
- **AI model interaction:** Python sends input to a model, runs inference, receives predictions or generated text, validates the output, and returns a response.

Example: Example: A Flutter app calls FastAPI. FastAPI authenticates the user, retrieves relevant documents, calls an LLM, stores the conversation, and returns the answer.

21. LLM vs vLLM: what is the difference?

Short Answer: An LLM is the language model itself. vLLM is an inference and serving engine designed to run LLMs efficiently, especially for high-throughput API serving.

Key Points:

- **LLM:** the model, such as a transformer-based language model.
- **vLLM:** software that loads and serves compatible models efficiently.
- vLLM can improve throughput and GPU utilization for production serving.

Example: Example: The model is the “brain”; vLLM is an optimized “server” that helps many users access that brain efficiently.

22. What is an AI Agent / Agentic AI?

Short Answer: An AI Agent is a system that can understand a goal, reason about steps, use tools, observe results, and continue until it completes the task.

Key Points:

- Goal → Plan → Tool → Observe → Decide → Repeat.
- Agents are useful when the task requires dynamic decisions instead of one simple LLM call.

Example: *Example: A travel agent can search flights, check hotel availability, compare prices, and prepare an itinerary.*

23. What are Multi-Agent Systems and CrewAI?

Short Answer: A Multi-Agent System uses multiple specialized AI agents that cooperate. CrewAI is a framework for defining agents, roles, tasks, and workflows.

Key Points:

- One agent may research.
- Another may analyze.
- Another may review or validate.
- A coordinator can combine the results.

Example: *Example: Research Agent → Data Analyst Agent → Reviewer Agent → Final Writer Agent.*

24. What is Generative AI?

Short Answer: Generative AI creates new content such as text, code, images, audio, or video by learning patterns from data.

Key Points:

- LLMs generate text and code.
- Diffusion models generate images.
- Audio models generate or transform speech and sound.

Example: Example: A user asks an AI system to create a product description from a few product details.

25. What are Tokens and Context Window?

Short Answer: A token is a small piece of text processed by an AI model. The context window is the maximum amount of input and output information the model can handle in one request.

Key Points:

- Longer text uses more tokens.
- More tokens can increase cost and latency.
- If context is too large, important information may be truncated or unavailable.

Example: Example: A RAG system retrieves only the most relevant document chunks so the LLM receives useful information within its context window.

26. What is Prompt Engineering?

Short Answer: Prompt Engineering is the practice of designing instructions and context so an AI model produces reliable and useful results.

Key Points:

- Clear role and task
- Relevant context
- Output format
- Examples when useful
- Constraints and validation

Example: Example: "Return JSON with fields name, price, and category. Do not add extra text." is more reliable than a vague instruction.

27. What is Fine-Tuning?

Short Answer: Fine-tuning trains a pre-trained model further on a specific dataset so it becomes better at a particular task or style.

Key Points:

- Use it when prompting or RAG is not enough.
- Requires good-quality data and evaluation.
- It changes model behavior; it does not automatically give the model current private knowledge.

Example: *Example: Fine-tune a model to follow a company's specific classification format or writing style.*

28. What is AutoGen?

Short Answer: AutoGen is a framework for building applications where AI agents and humans can communicate and collaborate to complete tasks.

Key Points:

- Agent conversations
- Tool use
- Multi-agent workflows
- Human-in-the-loop patterns

Example: *Example: A coding agent proposes a solution, a testing agent checks it, and a human approves the final change.*

29. LangChain vs LangGraph: why use both?

Short Answer: LangChain provides reusable components for LLM applications. LangGraph is designed for stateful, multi-step, branching, and cyclic agent workflows.

Key Points:

- **Use LangChain:** model calls, prompts, tools, retrievers, loaders, and chains.
- **Use LangGraph:** complex agent state, loops, branching, checkpoints, human approval, and multi-agent workflows.
- They can be used together: LangChain components inside a LangGraph workflow.

Example: Example: LangChain handles the retriever and LLM call; LangGraph controls the agent loop and decides whether another tool call is needed.



30. What are Llama and LlamaIndex?

Short Answer: Llama can refer to Meta's family of open language models. LlamaIndex is a framework for connecting LLMs with external data and building data-aware applications.

Key Points:

- **Llama model:** a language model that can be run through supported inference systems.
- **LlamaIndex:** document loading, indexing, retrieval, query engines, and agent/data workflows.

Example: Example: Load company PDFs → create an index → retrieve relevant sections → ask an LLM to answer from the retrieved data.

31. What is Reranking?

Short Answer: Reranking is a second-stage search step that reorders retrieved documents by relevance to the user's query.

Key Points:

- Vector search is fast and broad.
- A reranker is usually more precise but more expensive.
- The best documents are then sent to the LLM.

Example: *Example: Retrieve 50 possible documents, rerank them, and send the top 5 to the LLM.*

32. What is NLP?

Short Answer: Natural Language Processing (NLP) is the field of AI that helps computers understand, process, and generate human language.

Key Points:

- Text classification
- Translation
- Sentiment analysis
- Named entity recognition
- Question answering
- Summarization
- Speech-related language processing

Example: *Example: Extract a person's name, company, and job title from an unstructured CV.*

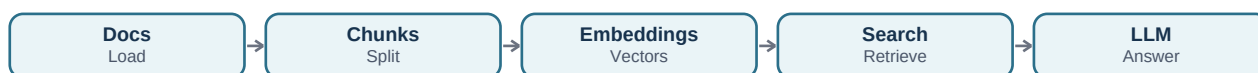
33. What is a RAG Pipeline?

Short Answer: Retrieval-Augmented Generation (RAG) gives an LLM relevant external information before it generates an answer.

Key Points:

- **Ingestion:** load documents.
- **Chunking:** split text.
- **Embedding:** convert chunks to vectors.
- **Indexing:** store vectors.
- **Retrieval:** find relevant chunks.
- **Reranking:** improve order.
- **Generation:** LLM answers using context.
- **Evaluation:** measure retrieval and answer quality.

Example: Example: A company chatbot answers HR questions using the company's latest policy documents instead of relying only on the model's training data.



34. What is a Vector Database?

Short Answer: A Vector Database stores numerical vector representations of data and searches for items that are semantically similar.

Key Points:

- Stores embeddings.
- Supports similarity search.
- Common use: RAG, recommendations, image search, and semantic search.

Example: Example: A question about “holiday policy” can find a document that says “annual leave rules” because their meanings are similar.

35. What is an Embedding?

Short Answer: An embedding converts text, images, audio, or other data into a list of numbers that represents its meaning or features.

Key Points:

- Similar meanings create vectors that are close in vector space.
- Embeddings are used for semantic search, recommendations, clustering, and RAG.

Example: Example: "How do I reset my password?" and "I forgot my login password" can have similar embeddings.

36. What are Semantic Search and Hybrid Search?

Short Answer: Semantic Search finds results by meaning using embeddings. Hybrid Search combines semantic search with keyword search.

Key Points:

- **Semantic:** good for meaning and paraphrases.
- **Keyword:** good for exact names, IDs, and technical terms.
- **Hybrid:** combines both for stronger retrieval.

Example: Example: Search for a product using both the exact SKU and the meaning of the user's description.

37. What are MCP Client, MCP Server, Tools, Resources, and Prompts?

Short Answer: MCP (Model Context Protocol) is a standard way for AI applications to connect models with external capabilities and context.

Key Points:

- **MCP Client:** the AI application that connects to servers.
- **MCP Server:** exposes capabilities or data.
- **Tools:** actions the model can request.
- **Resources:** readable context/data.
- **Prompts:** reusable instruction templates.

Example: Example: An AI coding client connects to an MCP server that exposes project files as resources and a test runner as a tool.

38. What are Plugins, and how can FastAPI/REST APIs be used with MCP or plugins?

Short Answer: A plugin is an extension that adds capabilities to an application. A FastAPI or REST API can become the backend behind a plugin or an MCP tool.

Key Points:

- **REST approach:** Plugin/MCP tool → HTTP request → FastAPI endpoint → business logic → database/service.
- **MCP approach:** MCP Server exposes a tool that internally calls your FastAPI or REST API.
- **Security:** authenticate, authorize, validate inputs, rate-limit, and never expose secrets to the model.

Example: Example: An MCP tool called `get_user_orders` calls `GET /orders` on your FastAPI backend and returns validated order data to the AI application.



39. How is Docker used in AI applications?

Short Answer: Docker packages an application, its dependencies, and runtime into a container so it behaves consistently across development, testing, and production.

Key Points:

- Package FastAPI and AI code.
- Run databases, Redis, and Vector DBs locally.
- Create reproducible development environments.
- Deploy inference services.
- Use Docker Compose for multiple services.
- Combine with Kubernetes for larger-scale orchestration.

Example: Example: One Docker container runs FastAPI, another runs Redis, another runs a Vector DB, and a model-serving container provides inference.



AI Engineering Architecture Cheat Sheet



Interview summary: Start from the user problem. Design the architecture. Choose the right model and data strategy. Build APIs and clients. Add security and evaluation. Deploy with containers and CI/CD. Monitor quality, latency, cost, and drift. Continuously improve the system.